7.5 Dilated convolution

- the repeated combination of max-pooling and striding at consecutive layers of these networks reduces significantly the spatial resolution of the resulting feature maps, typically by a factor of 32 across each direction
- a partial remedy is to use transposed convolution layers, which however requires additional memory and time
- instead, *dilated (atrous) convolution*, originally developed for the efficient computation of the wavelet transform in signal processing, can be used
- this algorithm allows us to compute the responses of any layer at any desirable resolution
- considering one-dimensional signals first, the output y_i of dilated convolution of a 1D input signal x_i with a filter w_k of length K is defined as:

$$y_i = \sum_{k=1}^{K} x_{i+r \cdot k} w_k.$$

- the rate parameter r corresponds to the stride with which we sample the input signal
- standard convolution is a special case for rate $r = 1$; see Figure 9 for illustration

7.5 Dilated convolution

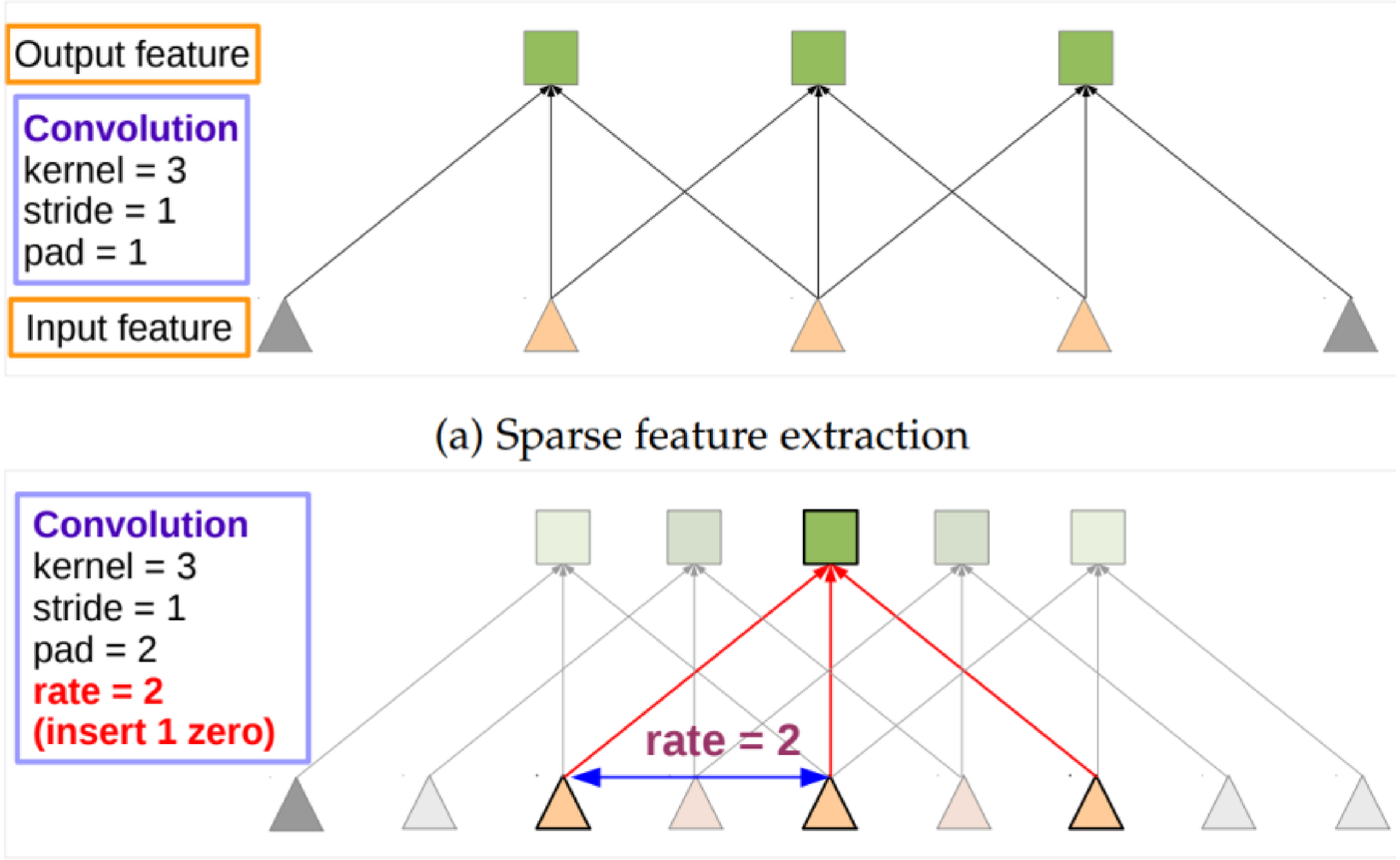


Figure 9: Illustration of dilated convolution in 1D. (a) Sparse feature extraction with standard convolution on a low-resolution input feature map. (b) Dense feature extraction with dilated convolution with rate $r = 2$, applied on a high-resolution input feature map.

7.5 Dilated convolution

- we illustrate the algorithm's operation in 2D through a simple example in Figure 10: given an image, we assume that we first have a downsampling operation that reduces the resolution by a factor of 2, and then perform a convolution with a kernel
- if we implant the resulting feature map in the original image coordinates, we realize that we have obtained responses at only $1/4$ of the image positions
- instead, we can compute responses at all image positions if we perform a convolution of the full-resolution image with a filter "with holes", in which we upsample the original filter by a factor of 2, and introduce zeros in between filter values
- although the effective filter size increases, we only need to take into account the non-zero filter values, hence both the number of filter parameters and the number of operations per position stay constant
- the resulting scheme allows us to easily and explicitly control the spatial resolution of neural network feature maps

7.5 Dilated convolution

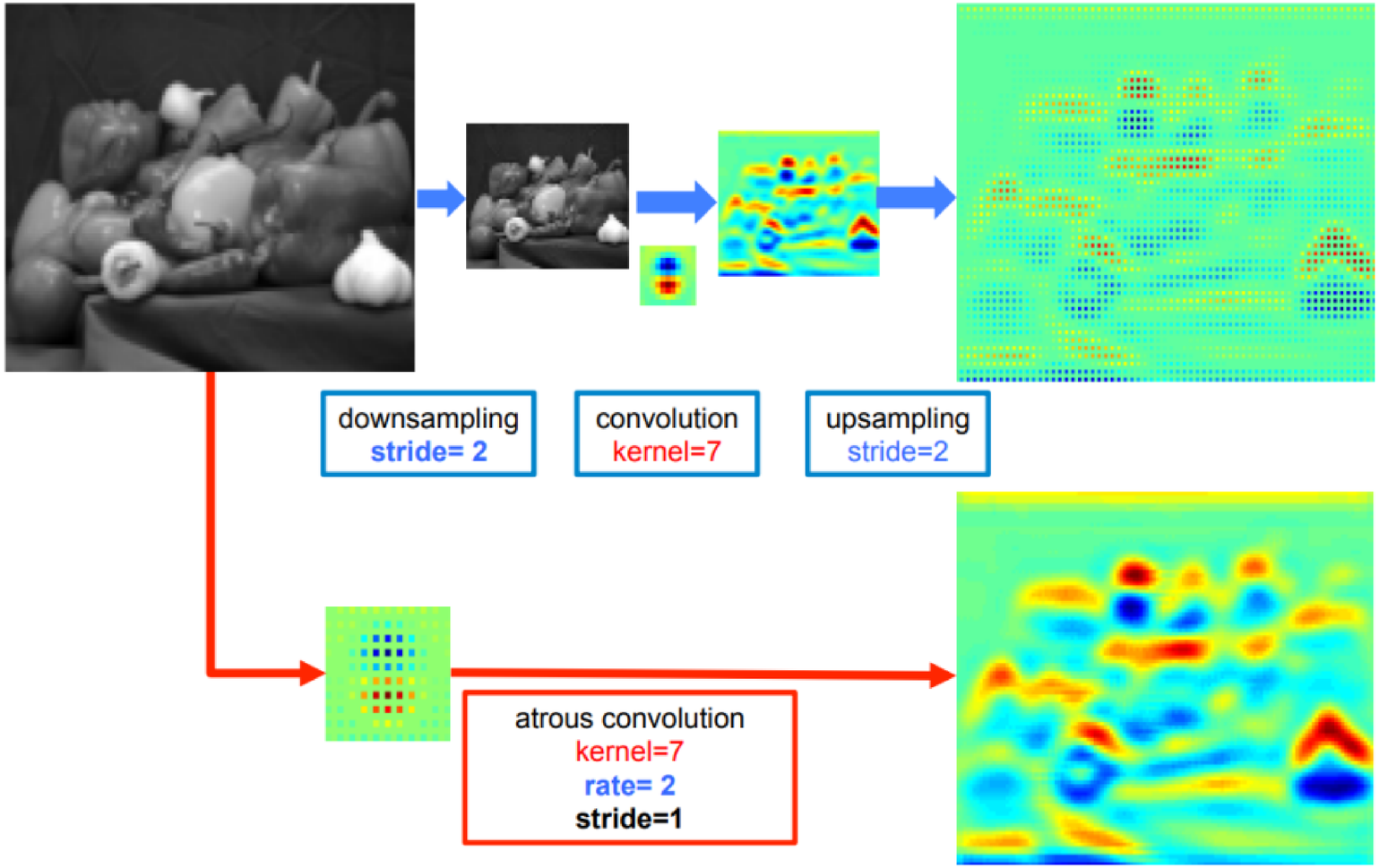


Figure 10: Illustration of dilated convolution in 2D. Top row: sparse feature extraction with standard convolution on a low-resolution input feature map. Bottom row: dense feature extraction with dilated convolution with rate $r = 2$, applied on a high-resolution input feature map.

7.5 Dilated convolution

- in the context of CNNs, we can use dilated convolution in a chain of layers, effectively allowing us to compute the final feature maps at an arbitrarily high resolution
- for example, in order to double the spatial density of computed feature maps in the VGG-16 or ResNet-101 networks, we find the last pooling or convolutional layer that decreases resolution, set its stride to 1 to avoid signal decimation, and replace all subsequent convolutional layers with atrous convolutional layers having rate $r = 2$
- pushing this approach all the way through the network could allow us to compute feature maps at the original image resolution, but this ends up being too costly
- a hybrid approach that has a good efficiency/accuracy trade-off, uses dilated convolution to increase by a factor of 4 the density of computed feature maps, followed by fast bilinear interpolation by an additional factor of 8 to recover feature maps at the original image resolution
- bilinear interpolation is sufficient in this setting because the class score maps (corresponding to probabilities) are quite smooth
- unlike the transposed convolution approach, this approach converts image classification networks into dense feature extractors without requiring learning any extra parameters, leading to faster training in practice

7.5 Dilated convolution

- dilated convolution also allows us to arbitrarily enlarge the receptive fields of filters at any layer
- CNNs typically employ spatially small convolution kernels (typically 3×3) in order to keep both computation and number of parameters contained
- dilated convolution with rate r introduces $r - 1$ zeros between consecutive filter values, effectively enlarging the kernel size of a $k \times k$ filter to $k_e = k + (k - 1)(r - 1)$ without increasing the number of parameters or the amount of computation
- it thus offers an efficient mechanism to control the receptive field and finds the best trade-off between accurate localization (small receptive field) and context assimilation (large receptive field), as shown in Figure 11
- in general, when the dilation for the height is r_h and the dilation for the width is r_w, the output shape is:

$$\lfloor (n_h - r_h(k_h - 1) - 1 + p_h + s_h)/s_h \rfloor \times \lfloor (n_w - r_w(k_w - 1) - 1 + p_w + s_w)/s_w \rfloor.$$

7.5 Dilated convolution

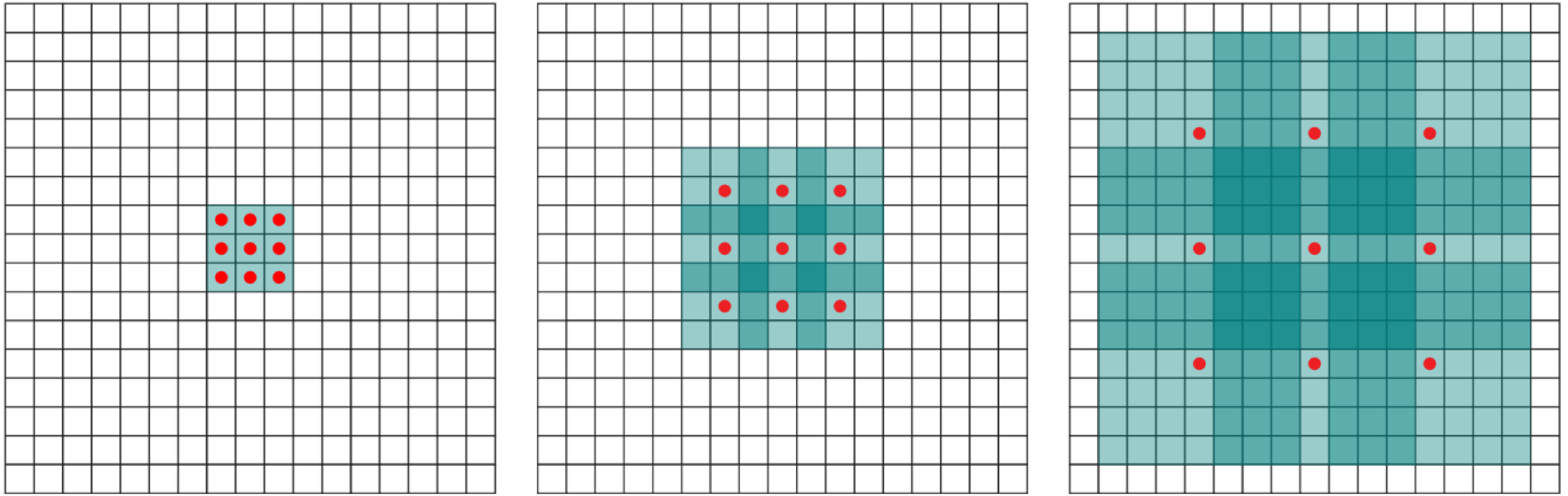


Figure 11: Systematic dilation supports exponential expansion of the receptive field without loss of resolution or coverage. (a) F_1 is produced from F_0 by a 1-dilated convolution; each element in F_1 has a receptive field of 3×3. (b) F_2 is produced from F_1 by a 2-dilated convolution; each element in F_2 has a receptive field of 7×7. (c) F_3 is produced from F_2 by a 4-dilated convolution; each element in F_3 has a receptive field of 15×15. The number of parameters associated with each layer is identical. The receptive field grows exponentially, while the number of parameters grows linearly.